Measuring Relationship Quality

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Abstract

This paper introduces a novel measure of relationship quality (RQ), the non-material gains from being in a relationship, and uses it to evaluate testable implications of different assumptions about match quality in models of couple formation and dissolution. We use survey data from the United Kingdom to construct and validate RQ, by demonstrating its predictive power for marital transitions and its comparability across partners. Exploiting the panel structure of the data, we test the empirical predictions of stochastic, learning and mixed models of match quality. We find that separation hazards decline with relationship tenure, the effect of negative RQ shocks diminishes over time, and shocks have a larger impact when occurring later in the relationship. These patterns support the stochastic framework and are inconsistent with learning models. These findings highlight the value of RQ for studying marital decisions and family stability.

Keywords: Match quality, marital decisions, divorce, measurement

JEL Codes: C81, J12, J13

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1 Introduction

Understanding the dynamics of marital transitions is a cornerstone of family economics. Decisions to form and dissolve couples shape household behaviour, individual well-being, and the effectiveness of public policy (Voena, 2015; Goussé et al., 2017). Research in family economics has emphasized the role of material gains from relationships, such as household specialization, economies of scale, and risk sharing, in sustaining couple stability (e.g., Becker, 1981). At the same time, this literature has also acknowledged the importance of non-material aspects of relationships (e.g., Browning et al., 2014).

Over recent decades, rising female labour force participation and the expansion of welfare states have weakened financial interdependence within couples, increasing the relative importance of non-material factors for couple formation and stability (Lundberg and Pollak, 2007). Despite this shift and even though theoretical work on couple formation and dissolution often considers it, empirical evidence on non-material gains from relationships, commonly referred to as match quality, remains scarce. As a result, key theoretical assumptions about the role and evolution of match quality remain largely untested. This restricts the value of such models to speak to couple decisions.

This paper addresses this gap by introducing a novel measure of the non-material gains from being in a relationship, which we refer to as relationship quality (RQ). We construct this measure using survey data and validate it by examining its theoretical properties and comparing it to prior empirical attempts. The measure allows us to test key implications of the different models of couple formation and dissolution, providing new empirical evidence on which assumptions are more accurate.

These models differ in their assumptions about match quality. One group of models treats match quality as an observed component of the utility function that follows a stochastic process, receiving shocks each period (e.g., Chiappori and Weiss, 2006; Voena, 2015; Greenwood et al., 2017). The second group treats match quality as an unobserved, deterministic component which is learned over time, with individuals receiving imperfect information and updating their beliefs about the match (Weiss and Willis, 1997; Brien et al., 2006; Antler et al., 2022).

Empirical to resolve these discrepancies are scarce and restricted, mainly due to data constraints. Within economics, these attempts often use observable characteristics of individuals or general well-being measures as proxies for match quality (Chiappori et al., 2018; Eckstein et al., 2019). A measure that explicitly incorporates relationship-specific information can provide a more accurate and reliable assessment of match quality (Bertrand et al., 2015). Outside of economics, several empirical studies have been conducted on marital satisfaction, marital conflict, and related concepts (e.g., Busby et al., 1995; Rijken and Liefbroer, 2009; Fernandes-Pires et al., 2023). However, they are constrained by the lack of longitudinal data, which is essential for testing the validity of different theoretical models.

We construct our measure of RQ using data from Understanding Society, a longitudinal survey dataset in the United Kingdom. This dataset includes a questionnaire containing ten questions related to relationships with cohabiting partners. These questions capture subjective assessments or personal evaluations of the relationship, such as the level of happiness with the relationship, and information on couple time use based on shared experiences, such as the number of joint activities. The combination of both blocks offers a nuanced view of the relationship. We use factor analysis to integrate all this information and construct RQ.

We conduct a series of tests to verify that our measure provides accurate information about couples' relationships. First, we assess the theoretical properties of informativeness and interpersonal comparability (Ferrer-i Carbonell and Frijters, 2004). Supporting informativeness, we find that RQ strongly predicts couple decisions, as periods before separation are characterized by significantly lower RQ, while RQ is higher before marriage and childbirth. In favour of interpersonal comparability, we find a high correlation in reported RQ across partners, with partner RQ being the most influential predictor of individual RQ.

We start to evaluate how prior literature has approached match quality by examining the associations between RQ and observable characteristics that have previously been used as proxies for match quality. Men, college-educated individuals, and married individuals report higher RQ on average, consistent with the patterns assumed in prior research, whereas age differences and income play only a minor role. Exploiting the panel structure of our data, we also examine the life-cycle and relationship profiles of RQ. We find that RQ is remarkably stable over time, as the positive association with age and the negative association with relationship tenure largely offset each other.

Importantly, we use RQ to evaluate the testable implications of different conceptual frameworks of match quality: the stochastic model, the learning model, and a mixed model that incorporates both mechanisms. Following Marinescu (2016), we focus on three specific outcomes: the slope of the separation hazard over relationship tenure, the persistence of negative match quality shocks on the hazard, and the effect of a quality shock on the hazard across tenure. A key advantage of our measure is that the panel structure of the data allows us to observe within-relationship, period-to-period changes in RQ. We define relatively large declines in RQ as negative shocks to match quality. This approach provides a more accurate analysis compared to previous work that relies on unemployment as a proxy for negative shocks, which has notable shortcomings.

We find that separation hazards decline with relationship tenure, that the effect of a negative RQ shock diminishes over time, and that shocks have a larger impact on separation when they occur later in the relationship. These patterns hold across different magnitudes of shocks, although they are more pronounced for large declines in RQ. Taken together, the evidence across these three outcomes is inconsistent with the predictions of learning models. The results are fully consistent with the stochastic model, while the findings do not allow us to definitively

rule out the mixed model.

These results guide future research in several ways. First, they identify which assumptions about match quality are supported by the data, providing a foundation for developing empirically grounded models to address key questions in family economics, such as how policies shape marital decisions. Second, the introduction of a novel and validated measure of match quality opens numerous avenues for empirical investigation. For instance, Rodríguez-Moro and Román (2023) use the longitudinal structure of RQ to examine the causal effect of childbirth on match quality and its interaction with household specialization. More broadly, RQ enables future research linking match quality to fertility choices, investments in children, and overall family stability, crucial insights for policymakers aiming to support families effectively.

Roadmap. The rest of the paper is organized as follows. Section 2 provides background on the theoretical foundations of match quality and reviews previous empirical attempts to measure it. Section 3 describes the data and explains the construction of RQ. Section 4 validates the measure. Section 5 examines its associations with observable characteristics and its evolution over time. Section 6 tests empirical implications of the different theoretical frameworks. Section 7 concludes.

2 Background

The concept of relationship or match quality is well-established in the social sciences. It refers to the non-material benefits derived from being in a relationship and it has been extensively explored in the theoretical family economics literature to understand family formation and dissolution decisions. At the same time, various social sciences, including economics, psychology, and sociology, have attempted to empirically measure this concept. In this section, we first outline the presence of relationship quality in economic theory, followed by a review of the different empirical measures proposed by and used across disciplines.

2.1 Theoretical Foundations of Match Quality

Understanding the formation and dissolution of couples, and broader family decisions, can be traced back to Becker (1981). Formally, there are two broad categories of gains that individuals derive from being in a couple: material and non-material gains. When making family decisions, individuals consider the trade-offs between these two types of gains (Charles and Stephens, 2004; Hess, 2004). Material gains relate to the factors that make being in a couple financially beneficial, such as economies of scale, household specialization, or risk sharing. In contrast, non-material gains are more subjective and include factors like love or companionship, which

are match-specific. Unlike material benefits, which can be obtained with any partner, non-material gains largely depend on the specific partner and the dynamics of the relationship.

Explicit efforts to account for the non-material component associated with relationships date back to Weiss and Willis (1997). The authors interpret match quality as characteristics of the couple or partner that minimize the probability of divorce and whose valuation may change over time. They distinguish between a permanent component, formed by observable characteristics, and a stochastic component, which is unobserved and determines the valuation of these traits over time.[1]

Building on this, part of the literature considers match quality to be a combination of observable characteristics and shocks to their valuation (Eckstein et al., 2019; Low, 2024). However, most papers have integrated this concept into family formation and dissolution models as a fully unobserved component (Browning et al., 2014; Chiappori and Mazzocco, 2017; Chiappori, 2020). These models can be divided into two groups based on their treatment of match quality.

The first group treats match quality as a stochastic component in the utility function. The parameter value is independent and identically distributed every period, conceiving realizations of match quality as non-persistent shocks (Chiappori and Weiss, 2006, 2007; Gemici and Laufer, 2011; Bruze et al., 2015; Voena, 2015; Goussé et al., 2017; Greenwood et al., 2017; Low et al., 2018). The decision to initiate or continue a relationship depends on whether the realization of the match quality parameter is above a certain threshold in a given period. This approach is the most prevalent in the literature.

The second group treats match quality as a deterministic, unobserved variable. In this framework, couple members receive a noisy signal about the true quality that is updated each period (Brien et al., 2006; Blasutto et al., 2020; Antler et al., 2022; Blasutto, 2024). A key distinction of this approach is the persistent nature of these signals, which allows couples to gradually learn and refine their understanding of *true* match quality as their relationship evolves.

However, the value of these models is limited without guidance on how match quality effectively determines couple decision making. To establish which treatment of match quality most accurately approximates reality, it is crucial to develop and test empirical proxies.

[1]Previous work accounted for related concepts related to match quality. Becker et al. (1977) considers observable characteristics of the partners contributing to marital gains, although these qualities are largely related to earnings potential. Lillard and Waite (1993) includes a stochastic residual component that accounts for observed heterogeneity in marital stability, after controlling for observable characteristics of the partners.

2.2 Empirical Studies of Match Quality

Empirical efforts within the economics literature are limited due to difficulties to measure this concept.[2] They primarily originate from studies relating match quality to observed partner characteristics. Weiss and Willis (1997) use a set of individual (education and age), couple (religious similarity and relationship type), and environmental characteristics (divorce law regimes) to proxy for the permanent component of match quality. Subsequent work has retained human capital variables and added assortative matching (Eckstein et al., 2019), *reproductive capital* (Low, 2024) and personality traits (Lundberg, 2012) to this measurement.[3]

Attempts to directly measure match quality are more scarce. Chiappori et al. (2018) infer match quality by trying to isolate the portion of subjective well-being that can be attributed to the relationship. However, this approach might be subject to measurement error due to unaccounted factors influencing well-being. Instead, measures explicitly integrating relationship-specific information can provide a more precise and reliable assessment in this context, shedding light on the evolution of match quality over the life cycle and in response to life events. Bertrand et al. (2015) advance in this regard using three questions related to happiness and conflict as separate measures of marital stability. While these measures come closer to capturing match quality, they focus on related but distinct aspects of relationships, leaving room for a more holistic approach.

In psychology, studies have explicitly tackled couple match quality by proposing various measures of marital satisfaction, conflict and related concepts. One of the first and most widely used measures is the Dynamic Adjustment Scale (DAS) proposed by Spanier (1976).[4] The DAS has been extensively tested and validated as a measure of marital satisfaction and stress (Norton, 1983; Graham et al., 2006; Funk and Rogge, 2007; Joel et al., 2020). It was later revised and condensed into a measure comprising responses to a 14-item questionnaire, divided into three subscales: consensus, cohesion, and satisfaction (Busby et al., 1995). In this paper, we propose a comprehensive measure of match quality that incorporates a subset of these items in a parsimonious way, integrating multiple aspects of relationships into a single metric.

Other disciplines have studied marital satisfaction and conflict in relation to a wide range of outcomes. Various findings reveal an association between these measures and marital transitions (Perelli-Harris and Blom, 2022), childbirth (Carlson and VanOrman, 2017; Rijken and

[2] In contrast, material benefits have been widely studied empirically in relation to marital stability, focusing on how changes in housing prices (Rainer and Smith, 2010; Farnham et al., 2011; Farzanegan and Gholipour, 2016), unemployment (Jensen and Smith, 1990; Charles and Stephens, 2004; Doiron and Mendolia, 2012; Eliason, 2012; González-Val and Marcén, 2017), and broader shocks to income (Becker et al., 1977; Charles and Stephens, 2004; Hess, 2004; Golosov et al., 2024) associate to couple dissolution.

[3] Kesternich et al. (2025) argue that relationship stability in the marriage and the labour market are associated with the same personality traits.

[4] Other seminal measures of marital satisfaction include the Marital Adjustment Test (Locke and Wallace, 1959), the Quality of Marriage Index (Hendrick, 1988), and the Relationship Assessment Scale (Norton, 1983), among others.

Liefbroer, 2009), physical and mental health (Holt-Lunstad et al., 2008; Meadows and Arber, 2015; Pieh et al., 2020; Fernandes-Pires et al., 2023), financial resources and labor market outcomes (Kinnunen and Feldt, 2004; Halliday Hardie and Lucas, 2010; Blom and Perelli-Harris, 2021), and overall happiness and well-being (Kamp Dush et al., 2008; Rhoades et al., 2009).[5]

However, previous analyses have some shortcomings, mostly coming from data restrictions. Past work is based on small, cross-sectional samples (Amato and Booth, 2001; Hassebrauck and Fehr, 2002), including match quality reports by non couple members or reports that rely on recall data (Feng et al., 1999; Amato and Patterson, 2017). The combination of a larger sample and longitudinal data associated with the measure constructed in this paper allows using previously unfeasible causal identification methods, providing more rigorous insights into these matters.

Additionally, our measure of match quality allows us to empirically assess the plausibility of the different models of couple formation and dissolution. Previous studies have attempted this in the context of couple dissolution. Marinescu (2016) explicitly tests both the stochastic and learning models outlined above. She develops different theoretical frameworks based on these models and derives testable predictions. Her findings suggest that learning plays a minor role, favoring the stochastic model. However, her use of job loss as a proxy for low match quality has limitations, since it primarily reflects material factors, rather than serving as an explicit measure of match quality.

3 Measure of Relationship Quality

3.1 Dataset and sample

We combine data from the British Household Panel Survey (BHPS) and Understanding Society (University of Essex and Research, 2024). The BHPS is a longitudinal household panel containing around 10,000 households and covering the period from 1991 to 2008. In 2009, it was replaced by Understanding Society, which includes 8,000 voluntary BHPS households and 40,000 new households. The survey is ongoing, with 14 waves released up to the year 2023.

The population of interest consists of adult individuals in cohabiting relationships, irrespective of their marital status. Accordingly, our sample includes respondents who are cohabiting with their partners and have available information on the outcome variable. The richness and longitudinal structure of the data enable us to construct the outcome for both partners and study its evolution across different ages and stages of the relationship. We restrict the sample to

[5] While most of these studies use measures based on a single question related to marital satisfaction, others adopt a more similar approach to ours by constructing multidimensional measures based on several items (Rijken and Liefbroer, 2009; Fernandes-Pires et al., 2023).

individuals aged 25-70 and couples with relationship tenures below 35 years.[6] The resulting sample is an unbalanced panel of 42,152 individuals observed up to six times.

3.2 Partner questionnaire

Every other wave of Understanding Society includes the Self-Completion Partner Module. This ten-item questionnaire is related to individuals' relationships and is completed individually by all respondents in cohabiting relationships, irrespective of their marital status. Eight of these items ask about the frequency of certain experiences within the couple, such as "How often do you and your partner calmly discuss something?" or "How often do you consider splitting?", which are answered on a six-point Likert scale.[7] The remaining two items ask respondents to express the degree of happiness with their relationship on a seven-point scale, and to report how many outside interests they engage in with their partners on a five-point scale. The full list of items is presented in Table 1.

This information is available for waves 1, 3, 5, 7, 9, 11 and 13 of Understanding Society, spanning the period from 2009 to 2022. However, an implementation error in wave 3 made the responses to "Do you and your partner engage in outside interests together?" incomparable with those from the other waves. Therefore, we exclude wave 3 and construct the RQ measure using the remaining waves.

Table 1: Questions in the Understanding Society Partner module.

(a) Subjective assessment	(b) Couple time use
How often do you... ?	*How often do you... ?*
discuss or consider splitting	work together on a project
regret that you married or lived together	have stimulating exchanges of ideas
quarrel	calmly discuss something
get on each other's nerves	kiss your partner
What is the... ?	*Do you and your partner... ?*
degree of happiness of your relationship	engage in outside interests together

Notes: This table displays the questions in the Understanding Society Self-Completion Partner Module. This questionnaire is included in waves 1, 3, 5, 7, 9, 11 and 13, spanning the period 2009-2022. We divide the items in (a) subjective assessments of the relationship and (b) couple time use, based on the information they convey. The first four questions in each block are responded in terms of frequency in a six-point Likert scale, the happiness question is answered in terms of degree in a seven-point scale and the responses to the outside interest question correspond to amounts in a five-point scale.

We categorize the items in Table 1 into two distinct blocks based on the nature of the information they provide. The first block contains items related to subjective assessments of the

[6] These restrictions are imposed to avoid outliers and data errors, as well as to ensure sufficient sample sizes in the analyses by age and tenure.

[7] While all frequency-based questions use a six-point Likert scale, the response categories differ across items. See Subsection A.1 for the exact wording and response options.

relationship, including questions regarding happiness and perceived conflict within the couple. These items are considered subjective because the concepts they address may not be uniformly interpreted by both partners, and the responses are provided on a less precise scale, as the perception of terms like "all the time" can vary between individuals. The second block includes items related to how partners spend their time together, such as the frequency with which they work together on projects or the number of joint activities they engage in. This block conveys relatively more objective information, with concepts that are more likely to be shared between partners and responses that are provided in more specific time units.

3.3 Construction of the measure

We construct the measure of relationship quality (RQ) using the ten items in Table 1. First, we transform the responses so that zero corresponds to the *worst* couple behaviours and habits, such as never working together on a project or considering divorce all of the time. This transformation ensures that all items are increasing in the quality of the relationship. We then conduct a factor analysis, detailed in Subsection A.2, on all ten items to identify latent factors that explain their shared variance. We retain the first factor as the RQ measure, which accounts for 40.49% of the total variation in the underlying items.[8]

Figure 1 displays the resulting factor loadings. All factor loadings are positive and larger than 0.5, indicating strong associations between the RQ measure and each item. The highest correlations with the RQ measure are found in items related to calmly discussing issues and regretting getting married, while the lowest correlations are with items related to the frequency of kisses and the number of outside activities partners engage in together.

3.4 Measure and sample descriptives

The resulting RQ measure is standardized to have mean zero and unit variance, with higher values indicating better relationship quality. Thus, RQ is expressed in units of standard deviations. Figure 2 displays the distribution of RQ separately for (a) women and (b) men. For both groups, the distribution is left-skewed, indicating that higher RQ levels are more frequently reported. Mean values are close to zero but slightly lower for women and higher for men, and dispersion is somewhat greater among women.

Table 2 reports descriptive statistics. Panel (a) summarizes individual characteristics for men and women separately.[9] In our sample, men are on average fifty years old, while women are approximately three years younger. More than a fourth of the respondents are college

[8] As shown in Figure A.1, the first factor has an eigenvalue of 4.05, which is more than three times larger than that of the next factor. The eigenvalue of the rest of the latent variables remains below one.

[9] Since we have a panel with varying observation frequencies for different individuals, we first compute individual averages and then summarize across all individuals.

Figure 1: Factor loadings of RQ.

Notes: This graph displays the factor loadings from different factor analyses on the items in the Understanding Society Partner module. Black circles correspond to the main RQ measure. Blue squares and orange triangles correspond to using men and women samples only. The eigenvalues are 4.05, 3.83 and 4.24, and the resulting factors explain 40.49%, 38.35% and 42.37% of the variation in the items, respectively.

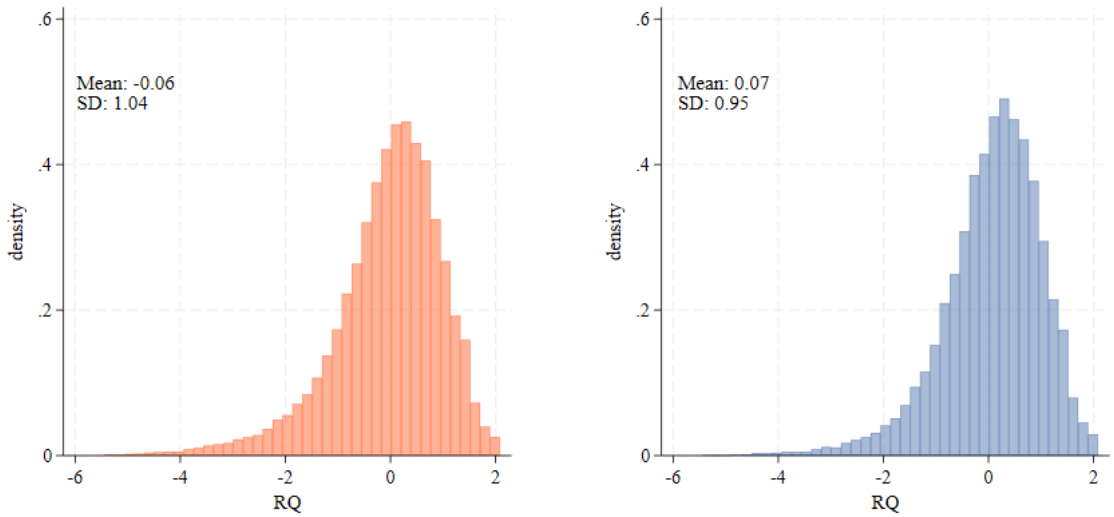
educated, and around 60% are active in the labour market and employed, with these numbers being lower for women. Men also work more hours and earn higher incomes than women, on average, while women spend over twice as much time as men on housework. Table 2 (b) summarizes characteristics at the couple level. About half of the couples in the sample are married, and the average relationship tenure is of 18 years. Finally, around a third of the couples have at least one child.

4 Measure Validation

Given the novelty of the RQ measure, we conduct a series of tests to verify that it provides meaningful and accurate information about relationship quality. First, we follow the approach in the life satisfaction literature and assess two key theoretical properties of the measure (Ferrer-i Carbonell and Frijters, 2004): informativeness and interpersonal comparability. We then compare our measure with previous empirical approaches discussed in Subsection 2.2.

Figure 2: Distribution of RQ.

(a) Women (b) Men



Notes: This figure displays the distribution of RQ separately for (a) women and (b) men. The measure is the first factor obtained by performing a factor analysis on the items in Table 1. It is standardized to have a mean of 0 and a standard deviation of 1.

4.1 Theoretical properties

Informativeness. We assess whether RQ provides meaningful information about the state of the relationship by examining its ability to predict major couple decisions: marriage, separation and fertility. Specifically, we compare the distribution of RQ in the period immediately before each event with the overall distribution.

Figure 3 plots the empirical cumulative distribution function (CDF) of RQ across these samples. Panel (a) compares the overall distribution of RQ with that observed one period before marriage and dissolution. The distribution before marriage is shifted to the right, indicating higher RQ among couples about to marry. This is consistent with marriage serving as a commitment mechanism that raises separation costs (Brien et al., 2006). In contrast, the distribution before separation is clearly shifted to the left. The shift before separation is notably larger, suggesting that couples often delay dissolution until RQ has deteriorated markedly, whereas smaller improvements are sufficient to trigger marriage. Panel (b) shows that among new parents, the RQ distribution is shifted upward before the birth of a first child, consistent with fertility being more likely in higher quality relationships. The Kolmogorov–Smirnov equality of distributions test reported in Table B.1 confirms that the pre-event distributions differ significantly from their benchmarks.

We further examine the correlation between RQ and marital transitions through regression analysis reported in Table B.2.[10] The outcome variable in the first two columns is a binary

[10]For the analysis on fertility, we refer to Rodríguez-Moro and Román (2023), which conducts a detailed study of the impact of becoming a parent on RQ.

Table 2: Summary statistics.

(a) Individual characteristics			(b) Couple characteristics	
	(1) Men	(2) Women		(3) Couples
Age	49.75	46.69	Tenure	17.93
	(16.11)	(15.68)		(15.47)
College educated (%)	27.12	25.85	Married (%)	49.42
	(43.58)	(42.62)		(47.84)
Active in labor mkt (%)	66.58	59.07	In urban areas (%)	76.45
	(38.40)	(39.89)		(41.25)
Employed (%)	62.81	56.11	Monthly household income	3835.4
	(39.00)	(40.39)		(6943.9)
Weekly work hours	21.18	15.51	At least one child (%)	32.03
	(16.88)	(13.96)		(42.80)
Gross monthly income	2267.8	1415.2		
	(1549.4)	(1104.9)		
Weekly housework hours	6.296	14.55		
	(5.238)	(8.458)		
RQ	0.0594	-0.0689		
	(0.894)	(0.999)		
Observations	19983	22168	Observations	46446

Notes: This table presents mean values of the set of (a) individual characteristics, separately for men and women, and (b) couple characteristics. Standard errors in parentheses.

indicator for the period preceding dissolution. A one–standard deviation increase in RQ is associated with a 0.6 percentage point lower probability of separation, a substantial association given the average annual dissolution rate of 2.5% reported in Figure B.1. The last two columns repeat the analysis for a binary indicator of the period preceding marriage. While RQ is positively correlated with marriage probabilities, the association becomes statistically insignificant once the full set of controls is included.[11]

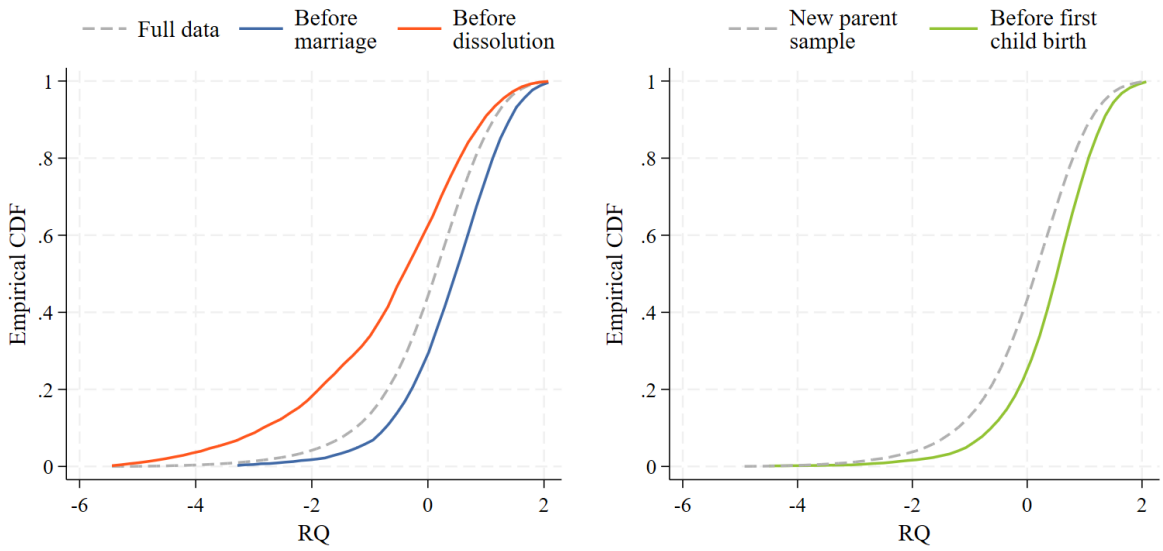
Interpersonal comparability. A valid measure of RQ should capture a concept that is at least partly shared between partners. We assess this by examining the correlation in RQ across couple members. We use a rank-rank plot to explore the non-linear relation between partners' RQ. Figure 4 displays the average RQ percentile rank of women by their partners' percentile rank. While the correlation is not perfect, which would produce a 45-degree line, there is a clear positive association between partners' RQ. The slope is steepest at the top and bottom of the distribution, approaching one, but flattens by nearly half in the middle ranks. This

[11]The specification controls for age, sex, college education, employment status, log monthly income, presence of children, relationship tenure, marital status, area of residence, and period.

Figure 3: Informativeness: behaviour prediction.

(a) Marital transitions

(b) First child birth



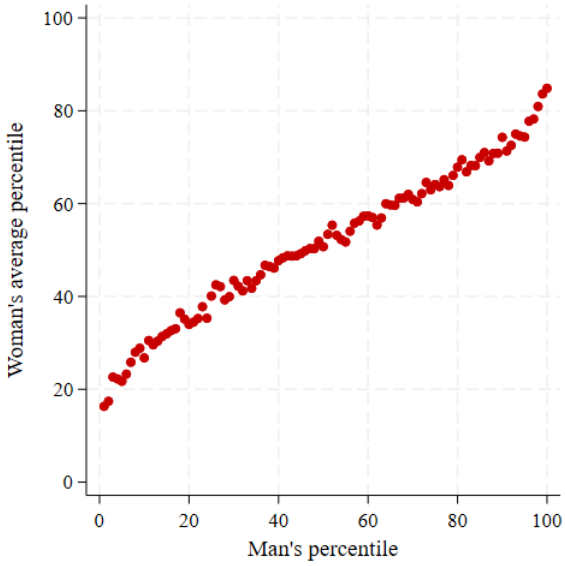
Notes: This figure displays the empirical cumulative distribution function (CDF) of RQ for different samples. Panel (a) compares the overall distribution of RQ in the full sample (dashed gray line) with its distribution one period before marriage (blue) and one period before couple dissolution (red). Panel (b) plots the overall distribution of RQ in the sample of individuals becoming parents (dashed gray line) and the distribution of the measure in this sample one period before the birth of their first child (green).

pattern suggests that extreme assessments of the quality of the relationship are more strongly shared between partners than intermediate ones. This pattern is sustained when controlling for observable characteristics in Figure B.2.

Table B.3 reports the results from regressing women's RQ on their (male) partners' RQ, indicating a linear correlation of 0.6 between partners' responses. Table B.4 additionally studies this correlation for each item in Table 1. There is a positive correlation in the responses to all items. Partners display the highest degree of agreement in the questions related to kissing and quarrelling, whereas the smallest correlation is given regarding stimulating exchanges of ideas and happiness with the relationship.

We also test for gender differences in how respondents value the items underlying the RQ measure by conducting factor analysis separately for men and women. Figure 1 plots the resulting loadings for the gender-specific factors. The male-specific factor exhibits lower loadings and a smaller eigenvalue than the pooled measure, while women's loadings are higher and the eigenvalue larger. Despite these differences in magnitude, the ranking of items is largely consistent across genders. The main difference is that for women the strongest loading corresponds to "regretting getting married", followed by "calmly discussing something", whereas for men the order of these two items is reversed.

Figure 4: Rank-rank correlation of RQ across couple members.



Notes: This figure presents the rank-rank correlation of RQ between couple members in the full data. We plot the average RQ percentile rank of women against their male partners' RQ percentile rank. Each dot represents a percentile.

4.2 Comparison with alternative measures

We study the correlation between our RQ measure and alternative measures of match quality proposed in the literature. As discussed in Section 2, several empirical attempts have proxied for match quality using observable characteristics of individuals or couples. We examine the association between RQ and observable characteristics in Section 5. Here we focus on explicit measures of match quality proposed in previous studies.

First, we consider the measure proposed by Chiappori et al. (2018), which is based on subjective well-being. The authors suggest that general well-being, measured using a life satisfaction scale, is the sum of marital quality, economic conditions and health. Their approach involves regressing overall life satisfaction on partner's life satisfaction, personal satisfaction with economic conditions and health status, own and partner's wages, age and time. This analysis is conducted separately for men and women, but accounts for interdependence by estimating the equations simultaneously. The residual from this regression is taken as the measure of relationship quality. We use Understanding Society data to derive an equivalent measure.[12]

Second, we examine the questions used by Bertrand et al. (2015) to measure marital stability. The first question asks respondents to describe their relationship on a seven-point happiness scale. We proxy this using the happiness item in Table 1, even if the response scales differ

[12] Although the survey questions we use are designed to capture the same concept as those used by Chiappori et al. (2018), they are not identical. While their satisfaction scales range from 1 (least satisfied) to 5 (most satisfied), our scales are more granular, ranging from 1 to 7. Additionally, our estimation approach differs: we estimate this in a single regression and account for correlation in the residuals by clustering at the couple level.

slightly. The other two questions are binary, asking whether the respondent felt their relationship was in trouble or discussed separation with their partner during the past year.[13] While we cannot proxy the trouble question, we can transform the item in Table 1 about the frequency of divorce discussions into a binary variable. This variable equals one when respondents report thinking about divorce "more often than not" or more frequently.

Third, we examine two sub-scales of the revised DAS proposed by Busby et al. (1995): cohesion and satisfaction. These sub-scales can be directly constructed using a subset of the questions listed in Table 1.[14] The Cohesion sub-scale sums responses related to the frequency of working together on a project, having stimulating exchanges of ideas, calmly discussing something and the number of shared outside interests. The Satisfaction sub-scale adds responses concerning the frequency of discussing divorce, regretting marriage, quarrelling and getting on each other's nerves.

We standardize all measures and compute their correlations with RQ. Table 3 presents the resulting correlation coefficients. The residualised subjective well-being measure has the lowest correlation with RQ. This measure relies on the assumption that all sources of life satisfaction other than marital satisfaction are accounted for in the residuals of overall well-being.[15] However, if other factors that could influence life satisfaction, such as having children or social life, are not fully accounted for, this could introduce measurement error. In contrast, the RQ measure specifically targets relationship quality with a partner, potentially reducing measurement error.

The questions used by Bertrand et al. (2015) are presented in (3) and (4). As expected, the correlation between RQ and couple happiness matches the factor loading of this item in the RQ measure. The question related to discussing separation is the only one negatively associated with RQ, which is reasonable given that discussions of separation indicate poorer relationship quality. This question is also part of the RQ measure, but assessed on a more granular scale. The advantage of RQ is that it integrates several aspects of couple relationships, like happiness, conflict or quality time together, into a unified measure, providing a comprehensive assessment of match quality.

The last two columns and rows of Table 3 correspond to the DAS sub-scales. Both sub-scales show a high correlation with RQ, being larger for satisfaction. This is expected since these sub-scales are constructed using a subset of the items used to create RQ. Additionally, the correlation between RQ and each sub-scale is higher than the correlation between the sub-scales themselves. Therefore, RQ effectively summarizes the information contained in the two

[13] Other than Bertrand et al. (2015), various studies use couple happiness and conflict as measures of marital satisfaction and stability (e.g., Halliday Hardie and Lucas, 2010; Perelli-Harris and Blom, 2022).

[14] There is a third sub-scale within the DAS, consensus, which we cannot construct. While the full DAS consists of the sum of the three sub-scales, each has also been studied independently.

[15] It also assumes additivity in how each of these factors contribute to well-being, excluding potential interaction effects.

sub-scales in a more parsimonious manner.

Table 3: Correlation among match quality measures.

	(1) RQ	(2) Subjective well-being	(3) Couple happiness	(4) Discuss separation	(5) DAS Cohesion	(6) DAS Satisfaction
(1) RQ	1					
(2) Subjective well-being	0.216	1				
(3) Couple happiness	0.672	0.186	1			
(4) Discuss separation	-0.338	-0.0547	-0.160	1		
(5) DAS Cohesion	0.808	0.159	0.413	-0.145	1	
(6) DAS Satisfaction	0.835	0.165	0.472	-0.431	0.411	1

Notes: This table reports the correlation coefficients among RQ and various match quality measures in the literature. Column and row (1) correspond to RQ. Column and row (2) present the residualised subjective well-being measure as proposed by Chiappori et al. (2018). Columns and rows (3) and (4) contain two of the marital stability measures used in Bertrand et al. (2015). Columns and rows (5) and (6) present the cohesion and satisfaction sub-scales of the revised DAS (Busby et al., 1995).

5 Basic Descriptives

5.1 RQ and observable characteristics

As discussed in Section 2, many empirical proxies for match quality in the economics literature are based on observable characteristics of individuals and couples. We investigate how RQ is associated with these factors by estimating a simple linear analysis where the outcome is the RQ of individual i at wave t. The results are displayed in Table 4. We gradually incorporate individual and couple characteristics into the model. Standard errors are clustered at the couple level.

We start by only including a set of individual characteristics: sex, college education, employment status, log monthly income and age. These were chosen following foundational family economics literature using observable characteristics to proxy match quality (Weiss and Willis, 1997) or exploring the link between marital stability and economic conditions (Becker, 1981; Marinescu, 2016). Column (1) presents the results of this estimation. Having a college degree is most strongly associated with RQ, with college educated individuals reporting 0.13 standard deviations higher RQ on average. Being employed also correlates with higher RQ values, while log monthly income shows a modest positive association. We find evidence that women systematically report lower levels of RQ compared to men.

As argued in Section 2, match quality significantly depends on the specific match, unlike material gains from a relationship. We examine how couples with different characteristics differ in RQ. These characteristics include marital status, urban residence, presence of children, age differences and relationship tenure. Column (2) presents the estimates with added couple

characteristics. The signs and significance levels of the individual characteristics remain largely unchanged, although magnitudes decrease, indicating some correlation between individual and couple characteristics. Two couple characteristics have much larger coefficients than any individual characteristic: married individuals report an RQ a fourth of a standard deviation higher than cohabiting couples, while individuals with at least one child report lower RQ. Urban residence and large age differences between partners are also associated with lower RQ.

Lastly, we add partner characteristics, which have also been proposed as measures of match quality, considering the same attributes as for individuals. Column (4) presents these estimates. Partner education and income have a stronger association with RQ than the individual's own education and income. Including partner education reduces the coefficient on individual education, revealing positive assortative matching on education. Partner employment, instead, is less relevant than own employment, but it remains significant.

The results so far highlights level differences across various individual and couple types. However, individual unobserved heterogeneity may correlate with both observables and RQ. To address this, we estimate the specification including individual fixed effects, removing unobserved individual heterogeneity and providing within-individual estimations. Thus, the resulting estimates refer to how changes in characteristics affect RQ for a given individual rather than doing comparisons across groups. Column (3) displays these estimates. All coefficients are smaller, and individual characteristics lose their significance, suggesting compositional differences between groups that correlate with RQ. Only the coefficients for children and marriage remain significant, though marriage is only marginally so. This indicates that RQ increases after marriage and that it decreases with the presence of children.

Gender differences. Previous research has considered the possibility that factors influencing match quality may vary by gender (Chiappori et al., 2018). To investigate this, we conduct the analysis above separately for men and women in Table C.1. The results reveal that the direction of these associations is largely consistent across genders. However, the coefficients for men are slightly higher for nearly all individual and couple characteristics. Notably, women's income shows a strong positive association with both their own and their partner's RQ, except when individual fixed effects are included. This result contrasts with Bertrand et al. (2015), who report a negative association between women's income and RQ, finding that when women become the primary earners, marital satisfaction decreases. They attribute this decline to challenges to traditional gender roles. However, it is important to note that our analysis does not consider relative income within couples and that most women in our sample are secondary earners. Additionally, Chiappori et al. (2018) suggest that the relationship between income and match quality may be non-linear. While small positive income shocks enhance RQ for both genders, larger shocks increase divorce probabilities by improving an individual's outside options.

Table 4: Regression of RQ on individual and couple characteristics.

	(1)	(2)	(3)	(4)
(a) Individual characteristics				
Female	-0.1066***	-0.0986***	-0.0912***	
	(0.0090)	(0.0090)	(0.0091)	
College	0.1195***	0.0981***	0.0628***	0.0026
	(0.0133)	(0.0132)	(0.0129)	(0.0379)
Employed	0.1321***	0.1049***	0.0850***	0.0321*
	(0.0169)	(0.0168)	(0.0172)	(0.0169)
Log monthly income	0.0088*	0.0071	0.0098*	-0.0019
	(0.0052)	(0.0051)	(0.0055)	(0.0047)
(b) Couple characteristics				
Married		0.2595***	0.2468***	0.0936***
		(0.0187)	(0.0227)	(0.0271)
Children		-0.2466***	-0.2537***	-0.0557***
		(0.0150)	(0.0169)	(0.0166)
Urban		-0.0665***	-0.0628***	-0.0051
		(0.0148)	(0.0166)	(0.0305)
Age difference ≥ 5		-0.0565***	-0.0148	0.0922
		(0.0140)	(0.0169)	(0.0998)
(c) Partner characteristics				
College			0.0808***	0.0375
			(0.0127)	(0.0353)
Employed			0.0414***	0.0052
			(0.0158)	(0.0158)
Log monthly income			0.0158***	-0.0004
			(0.0054)	(0.0046)
Age and wave	✓	✓	✓	✓
Tenure		✓	✓	✓
Individual FE				✓
R-squared	0.028	0.052	0.054	0.080
Observations	68617	68371	54160	54160

Notes: This table presents the results from regressing RQ on a set of individual, couple, and partner characteristics. Column (1) includes individual characteristics: age, gender, college education, employment, log monthly gross personal income, age and period. In column (2), we add couple characteristics: marital status, presence of children, urban residence, age difference and tenure. Column (3) additionally includes partner characteristics: college education, employment and log monthly income. Column (4) adds individual fixed effects. Standard errors clustered at the couple level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Assortative matching. Assortative matching on education has been shown to importantly influence couple formation and dissolution, as well as the way in which intra-couple decisions are made (Chiappori et al., 2017). In this line, Eckstein et al. (2019) consider assortative matching on education to be part of the match quality of a couple. Table 4 provides evidence of assortative matching is positively associated with RQ. To explore this further, we categorize couples

into four groups based on their educational attainment (Costa and Kahn, 2000). The results, in Table C.3 indicate that couples where none are college educated have the lowest RQ levels, and couples where both are college educated report the highest RQ. This result suggests that the positive association between education and RQ may reflect unobserved characteristics shared by college-educated individuals that contribute to better relationships, such as personality traits (Dupuy and Galichon, 2014) or gender norms (Blasutto, 2024; Bertrand et al., 2015).

5.2 RQ over time

We study how RQ evolves over time computing its life-cycle and relationship profiles by estimating the following regression model:

$$y_{i,t} = \alpha_i + \mu_t + \sum_{a=25}^{70} \mathbb{1}\{\text{age}_{i,t} = a\}\gamma_a + \sum_{d=1}^{35} \mathbb{1}\{\text{tenure}_{i,t} = d\}\delta_d + u_{i,t} \tag{1}$$

where $y_{i,t}$ refers to the RQ of individual i at wave t. The terms $\mathbb{1}\{\text{age}_{i,t} = a\}$ and $\mathbb{1}\{\text{tenure}_{i,t} = d\}$ denote the full set age and tenure dummies, respectively, and provide the life-cycle and relationship profiles of RQ. The model also includes individual and period fixed effects, α_i and μ_t, to isolate the age and tenure profiles of RQ by removing individual unobserved heterogeneity. Standard errors are clustered at the couple level.

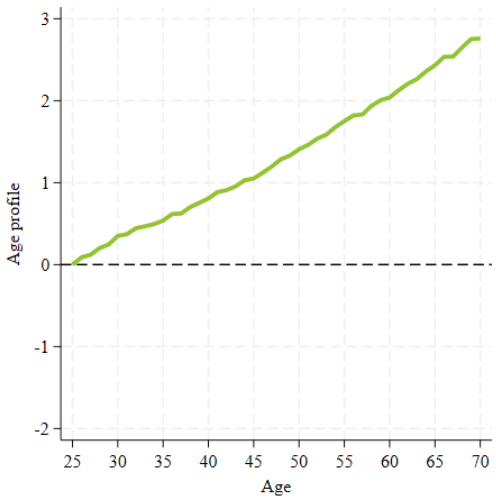
Figure 5 (a) illustrates the age profile of RQ, plotting the estimates of γ_a for all considered ages, taking 25 as the baseline. RQ increases almost linearly with age, without noticeable jumps.[16] Similarly, (b) shows the evolution of RQ with relationship tenure, using one year relationships as the baseline. Unlike the age profile, the tenure profile indicates a decreasing trend in RQ, with a slight convex shape where the decline becomes less steep for longer relationships. These findings indicate that age and tenure have opposing effects on RQ. Older individuals report higher RQ levels regardless of relationship tenure, while RQ generally decreases as relationship tenure increases.

To account for this and the possibility that the valuation of individual characteristics evolves over time (e.g., Becker et al., 1977), we compute the age and tenure profiles conditioning on the observables from Subsection 5.1 in Figure C.1. The results remain largely consistent, with slightly larger magnitudes for tenure. We also compute the age and tenure profiles separately. The age profile flattens out when considered in isolation, remaining flat until around age 45, when it increases linearly but less steeply than in Figure 5. The tenure profile, instead, remains largely unchanged, emphasizing the significance of tenure in determining RQ.

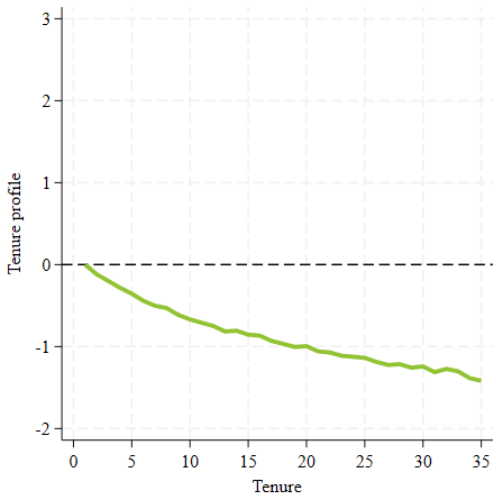
[16] Note that Equation 1 is a non-parametric function of age and tenure, allowing full flexibility in the relation between RQ and age, which makes the linear pattern more remarkable.

Figure 5: Age and tenure profiles of RQ.

(a) Age effects



(b) Tenure effects



Notes: This figure plots the (a) age and (b) tenure profiles obtained from estimating Equation 1. (a) plots the estimates of γ_a using 25 year-olds as a baseline, and (b) plots the estimates of δ_d using one year long relationships as a baseline.

6 Duration

In this section, we use our empirical measure of RQ to test the assumptions embedded in the leading theoretical models of couple formation and dissolution. As discussed in Subsection 2.1, the literature typically distinguishes two conceptual frameworks for the evolution of match quality throughout relationships: models of stochastic match quality and learning models. Each framework yields distinct predictions for the trajectory of separation hazards over relationship tenure and for how these hazards respond to shocks in match quality.

This analysis builds on Marinescu (2016), which presents simple versions of the stochastic and learning models, and a mixed model that incorporates both mechanisms. That paper derives testable implications for three key outcomes related to the separation process. The predictions of each framework are summarized in Table 5. The first outcome concerns the slope of the separation hazard over relationship tenure. The stochastic model predicts a decreasing hazard, while both the learning and mixed models imply a hazard that initially increases and then decreases. The second outcome relates to the persistence of a match quality shock on the hazard. Under the stochastic and mixed models, a shock initially raises the probability of separation, but the effect converges to zero over time. The learning model predicts an effect that increases with time since the shock. The third outcome evaluates how the impact of a shock varies with tenure at the time of its occurrence. The stochastic model predicts a larger effect at longer tenures, the learning model predicts a smaller effect, and the mixed model implies a non-monotonic pattern in which the effect first decreases and then increases with tenure.

Our empirical approach follows Marinescu (2016) but has one key advantage. The reference study relies on unemployment spells as proxies for negative shocks to match quality, which has a number of shortcomings. Most notably, the paper lacks a direct measure of match quality, so it cannot verify that unemployment is associated with a decrease in match quality. In fact, Rodríguez-Moro and Román (2023) show that becoming unemployed is not associated with a significant drop in RQ. In contrast, we exploit the availability of a direct measure of match quality and the longitudinal structure of our data. This allows us to observe period-to-period changes in RQ and to identify both the timing and the magnitude of within-relationship declines in match quality.

We define shocks using the distribution of period-to-period changes in RQ and consider two magnitudes. Declines of at least 0.5 standard deviations are classified as *small shocks*, and declines of more than one standard deviation are classified as *big shocks*. For each shock type, we retain only the first occurrence. Subsection D.1 provides a detailed description of these shocks in our data. Around 62% of couples in our sample experience at least a small shock and 25% experience at least one big shock. Most couples experience a single shock, and the share and magnitude of shocks are remarkably stable over tenure.

Table 5: Testable predictions under alternative conceptual frameworks of match quality.

		Prediction under different models		
		Stochastic	Learning	Mixed
(1)	Slope of the separation hazard with relationship tenure	Decreasing	Increasing, then decreasing	Increasing, then decreasing
(2)	Persistence of the effect of a match quality shock on the hazard	Positive initial effect, converging to zero over time	Increasing	Positive initial effect, converging to zero over time
(3)	Effect of a match quality shock on the hazard with tenure at shock	Larger effect at longer tenures	Smaller effect at longer tenures	Decreasing, then increasing with tenure

Notes: This table summarizes the empirically testable predictions implied by the three conceptual frameworks of match quality discussed in Marinescu (2016). The first column describes each testable outcome, while the remaining columns report the corresponding predictions under the stochastic, learning and mixed models of match quality.

6.1 Empirical strategy

To test these predictions, we estimate Cox proportional hazard models of couple separation. The hazard rate of separation of individual i in couple c at relationship tenure t is assumed to take the following form:

$$\lambda(t \mid X_{i,c}) = \lambda_0(t) \exp\left\{\beta' X_{i,c} + \gamma(t)\,\text{Shock}_{c,t}\right\}; \quad \text{where} \quad \gamma(t) = \gamma_0 + \gamma_1 t, \tag{2}$$

where $\text{Shock}_{c,t}$ is a binary variable equal to one if couple c has experienced a negative match quality shock by tenure t. To study how the effect of a shock evolves with time since the event, we alternatively replace this variable with an indicator equal to one if the couple experienced

a shock more than five years earlier. This specification relaxes the proportional hazards assumption by allowing the effect of a shock to vary with relationship tenure through $\gamma(t)$, which allows us to examine how the impact of a shock depends on the tenure at which it occurs.

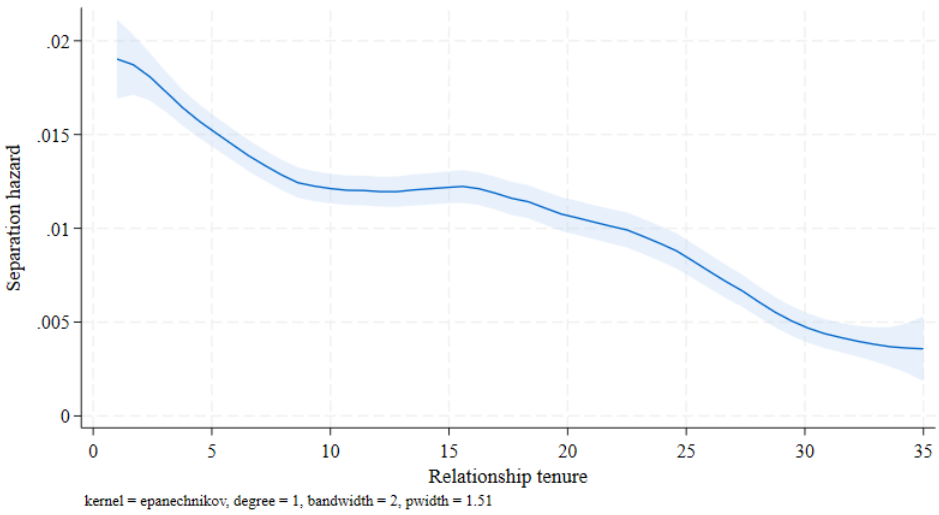
The vector $X_{i,c}$ includes time-invariant individual and couple level controls: an indicator for whether an individual had been married prior to the current relationship, and a dummy equal to one when the age difference between partners exceeds five years. In all specifications, we stratify the baseline hazard by sex, by the educational attainment of both partners, and by whether the couple eventually marries or has children at any point during the relationship.

6.2 Results

(1) Slope of the separation hazard with relationship tenure

Figure 6 displays the association between the probability of couple separation and relationship tenure, or the separation hazard. We use a local polynomial with an Epanechnikov kernel, degree one, and a bandwidth of two. This approach yields a flexible approximation of how separation risks vary with duration without imposing parametric structure. The hazard is decreasing with tenure. The probability of separation falls throughout the first ten years of a relationship, stabilizes over the next few years, and resumes a gradual downward trend. This pattern is consistent with the predictions of the stochastic match-quality model, which implies a monotonically decreasing separation hazard over relationship duration.

Figure 6: Separation hazard.



Notes: This figure plots the hazard function of couple separation along relationship tenure. This association is obtained using a local polynomial with an Epanechnikov kernel, degree one, and a bandwidth of two, providing an approximation that does not rely on parametric assumptions about the underlying association.

(2) Persistence of the effect of a match quality shock on the hazard

To examine how the effect of a match quality shock evolves with time since the shock, we estimate Equation 2 using two alternative definitions of the $\text{Shock}_{c,t}$ variable. In the *short-run* specification the indicator equals one if the shock occurred within the past five years, while in the *long-run* specification it equals one if the shock occurred more than five years ago. Comparing the coefficients across these two specifications provides evidence on the persistence of the effect: if the estimated coefficient is larger in the short run, the impact of the shock diminishes with time since the event.

Table 6 reports the corresponding results. The first row presents estimates for the short-run specification and the second for the long-run specification. Columns (1) and (2) consider the small shock, while columns (3) and (4) use the big shock definition. Across all specifications, match quality shocks are associated with statistically significant increases in separation probabilities. As expected, the magnitude of this association is larger for big shocks than for small shocks, indicating that separation risks respond proportionally to the severity of the decline in RQ. When controls are added, effect sizes increase slightly, although there is a loss of precision.

Crucially, the coefficients are consistently larger in the short-run specification than in the long-run specification, being almost double. In other words, among couples who experience a decline in match quality, those who do not separate within the first five years after the shock face substantially lower subsequent separation risks. This pattern is inconsistent with the predictions of learning models, which imply that the impact of a negative realization should grow over time as partners update their beliefs. Instead, the evidence is consistent with the stochastic and mixed models. In these frameworks, where match quality evolves through transitory shocks, a decline in match quality should increase the probability of separation, but its effect should be short-lived. While we do not observe the effect fully dissipating, its magnitude clearly diminishes over time.

(3) Effect of a match quality shock on the hazard over tenure

Finally, to study how the effect of a match quality shock varies with relationship tenure, we relax the proportional hazards assumption as specified in Equation 2. This specification allows the effect of the shock to change with tenure by estimating both γ_0 and γ_1. The coefficient γ_0 provides the baseline level effect of the shock, while γ_1 measures how this slope varies with tenure, that is, it corresponds to the interaction between the shock and tenure.

Table 7 presents the results. The estimated level effects are smaller than those in Table 6, where we constrained the effect of the shock to be constant over tenure. This suggests that part of the total impact of a shock can be attributed to changes over tenure. As before, the level coefficient is larger for big shocks than for small shocks, and the magnitudes of the coefficients

Table 6: Impact of match quality shocks on separation as a function of time since shock.

	Small shock (1/2 SD)		Big shock (1 SD)	
	(1)	(2)	(3)	(4)
≤ 5 years since shock	0.438***	0.719**	0.761**	1.008
	(0.038)	(0.074)	(0.078)	(0.116)
> 5 years since shock	0.282***	0.410***	0.440**	0.596*
	(0.050)	(0.070)	(0.117)	(0.123)
Controls		✓		✓
Observations	35444	30172	35444	30172

Notes: This table presents the results from estimating Equation 2 separately for shocks that occurred less than five years ago (first row) and for shocks taking place more than five years ago (second row). Columns (1) and (2) consider small match quality shocks (more than half a standard deviation of period-to-period RQ changes), whereas columns (3) and (4) use big shocks (more than one standard deviation). Controls in columns (2) and (4) include an indicator for whether an individual had been married prior to the current relationship, and a dummy equal to one when the age difference between partners exceeds five years. In those specifications we also stratify the baseline hazard by sex, by the educational attainment of both partners, and by whether the couple eventually marries or has children at any point during the relationship. Standard errors clustered at the couple level in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

slightly increase when including controls and stratifying.

The time-varying coefficients, or the interactions between the shock and tenure, are sizeable and statistically significant. For small shocks, the estimated association with separation increases by roughly 12 percent per additional year of tenure. For large shocks, the corresponding increase is about 7 percent. These estimates indicate that the impact of a match quality shock intensifies as relationships progress. This pattern cannot be reconciled with the learning model, which predicts that the effect of a negative realization should diminish at longer tenures. Instead, it is consistent with the stochastic match quality framework.

Taken together, the evidence across the three testable implications proposed by Marinescu (2016) is not consistent with the predictions of the learning model outlined in Table 5. Each result aligns with the predictions of the stochastic model, which implies decreasing separation hazards with tenure, shock effects that dissipate with time since occurrence, and larger shock effects at longer tenures. While the findings do not allow us to definitively rule out the mixed model, they provide no empirical support for the learning model.

7 Concluding Remarks

In this paper, we introduce a novel measure of relationship quality, a proxy for the non-material benefits of being in a relationship. This measure addresses a long-standing gap in the literature, which has primarily focused on match quality from a theoretical perspective, often neglecting empirical measurement and practical guidance. Our measure is built upon multidimensional

Table 7: Impact of match quality shocks on separation probabilities over relationship tenure.

	Small shock (1/2 SD)		Big shock (1 SD)	
	(1)	(2)	(3)	(4)
Shock	0.157***	0.272***	0.403***	0.611**
	(0.020)	(0.045)	(0.054)	(0.108)
Shock × Tenure	1.126***	1.095***	1.070***	1.045**
	(0.016)	(0.014)	(0.013)	(0.014)
Controls		✓		✓
Observations	35444	30172	35444	30172

Notes: This table presents the results from estimating Equation 2 relaxing the proportional hazards assumption and allowing for time-varying coefficients. Columns (1) and (2) consider small match quality shocks (more than half a standard deviation of period-to-period RQ changes), whereas columns (3) and (4) use big shocks (more than one standard deviation). Controls in columns (2) and (4) include an indicator for whether an individual had been married prior to the current relationship, and a dummy equal to one when the age difference between partners exceeds five years. In those specifications we also stratify the baseline hazard by sex, by the educational attainment of both partners, and by whether the couple eventually marries or has children at any point during the relationship. Standard errors clustered at the couple level in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

and longitudinal data, capturing various aspects of relationships in a parsimonious way and allowing us to observe the same relationship at different points in time. Through rigorous validation, we show that our measure effectively captures the intended concept. It predicts marital transitions, accurately reflecting the real status of relationships, and exhibits a significant within-couple correlation, indicating a shared perception of relationship quality.

The introduction of this measure opens new avenues for research, some of which are explored in this paper. Notably, the longitudinal nature of our data allows us to test assumptions from existing models about the evolution of match quality. Our findings support the view of match quality as a stochastic process, where partners receive match quality shocks each period. Additionally, this measure enables studying other theoretical assumptions, such as the additive nature of material and non-material relationship gains, or whether match quality is gender-specific, which has important implications on the transferable utility assumption.

Having an empirical measure for a previously unquantified concept enables analyses that were not possible before and offers new insights into policy discussions on marriage, divorce and fertility. For instance, Rodríguez-Moro and Román (2023) use the RQ measure to demonstrate the lasting negative impact of first childbirth on relationship quality, mediated by unequal housework arrangements. Such analyses deepen our understanding of declining fertility rates and have the potential to inform policy debates on parental leave policies and paternity support systems.

Future research could explore additional issues, such as the implications of parental relationships and their dissolution on children. While extensive research has been conducted on the effects of parental divorce on child outcomes, these studies often lack a proper counterfac-

tual scenario that considers what might have occurred if the relationship had remained intact. This new measure allows us to compare unhappy parents who remain together with those who separate, providing valuable and novel insights for divorce policies.

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A Measure Construction

A.1 Self-Completion Partner Module

Text: *For each of the following questions, please indicate which best describes your relationship with your partner at the moment. Please select only one answer per question.*

How often do you have a stimulating exchange of ideas?

1	Never
2	Less than once a month
3	Once or twice a month
4	Once or twice a week
5	Once a day
6	More often

How often do you calmly discuss something?

1	Never
2	Less than once a month
3	Once or twice a month
4	Once or twice a week
5	Once a day
6	More often

How often do you work together on a project?

1	Never
2	Less than once a month
3	Once or twice a month
4	Once or twice a week
5	Once a day
6	More often

How often do you discuss or consider divorce, separation or terminating your relationship?

1	All of the time
2	Most of the time
3	More often than not
4	Occasionally
5	Rarely
6	Never

Do you ever regret that you married or lived together?

1	All of the time
2	Most of the time
3	More often than not
4	Occasionally
5	Rarely
6	Never

How often do you and your partner quarrel?

1	All of the time
2	Most of the time
3	More often than not
4	Occasionally
5	Rarely
6	Never

How often do you and your partner "get on each other's nerves"?

1	All of the time
2	Most of the time
3	More often than not
4	Occasionally
5	Rarely
6	Never

Do you kiss your partner?

1	All of the time
2	Most of the time
3	More often than not
4	Occasionally
5	Rarely
6	Never

Do you and your partner engage in outside interests together?

1	All of them
2	Most of them
3	Some of them
4	Very few of them
5	None of them

The responses below represent different degrees of happiness in your relationship. The middle point, "happy", represents the degree of happiness of most relationships. Please se-

lect the number which best describes the degree of happiness, all things considered, of your relationship.

1	Extremely unhappy
2	Fairly unhappy
3	A little unhappy
4	Happy
5	Very happy
6	Extremely happy
7	Perfect

Responding to the module. Due to the personal nature of the questionnaire, preserving the privacy of respondents is a priority of Understanding Society. The questionnaire is part of a self-completion module, meaning that it is not administered by an interviewer but completed individually, either on paper or a computer, regardless of the interview mode.[17]

Although the questionnaire is answered individually, respondents may still be concerned about others being present during the interview. Individuals are given the option to refuse to answer the self-completion part of the survey, and if they do so due to the presence of another person, this reason is recorded separately. However, this is rarely reported: in the most recent wave, no one mentioned it, and in the previous wave, only three out of 363 refusals were due to another person being present.[18] Additionally, when an interviewer is present, they record whether another person is present and whether that person influences the interview. In the majority of cases, no one else is present during the interview.

A.2 Factor Analysis

Factor analysis is a statistical dimension reduction method that seeks to disentangle latent associations between different items. The goal is to identify a set of latent variables, or factors, that explain the shared variance among a set of observed items. The variance of each item is assumed to be influenced by these common factors, as well as by a unique error term specific to each item. The factors, which are unobservable, are inferred from their effects on the items. They are identified by solving a system of equations that produces a set of weights, called factor loadings, which indicate how strongly each item is associated with each factor.

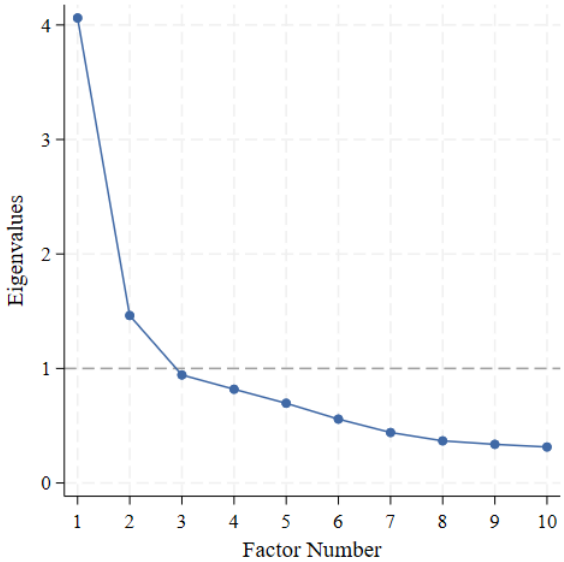
Crucially, factor analysis differs from principal component analysis (PCA), another common data reduction method, in both assumptions and interpretation. PCA aims to reduce dimen-

[17]Understanding Society conducts interviews in three modes: face-to-face, by phone, and online. Until wave six, all interviews were conducted either face-to-face (98%) or by phone. Web mode was introduced in wave seven, and its use has steadily increased, reaching 87% in the most recent wave (13).

[18]Other reasons for refusal include disliking the computer format, a child needing attention, confidentiality concerns, lack of motivation, or time constraints.

sionality by finding the linear combinations of items that capture the largest possible variance, driven purely by the data and without assuming an underlying structure. In contrast, factor analysis assumes that observed variables are influenced by latent factors and explicitly models the unique variance of each variable. Its goal is to uncover the latent constructs rather than simply reducing the number of dimensions. In our case, the responses to the questionnaire items in Table 1 are the observed variables. Instead of treating relationship quality as a linear combination of the items, as PCA would, we consider that this variable is a latent factor influencing respondents' answers to each item of the questionnaire.

Figure A.1: Eigenvalues from factor analysis.



Notes: This graph presents the eigenvalues of all factors derived from performing a factor analysis on the responses to the ten questions in the Understanding Society Partner module. The eigenvalues indicate the amount of variation in the sample explained by each factor. The first factor from this analysis is the RQ measure.

Before conducting the factor analysis, we recode the responses to the items in Subsection A.1 so that zero represents the least favourable couple behaviours and habits, like constantly quarrelling or never kissing one's partner, to ensure that all item responses are increasing in the quality of the relationship. We then perform factor analysis on all available responses to the ten items, and retain the first factor as the RQ measure. The decision to retain a single factor is based on the eigenvalues plotted in Figure A.1. The first factor has an eigenvalue of 4.06, more than twice that of the second next factor, indicating a substantial drop in the explained variation. All subsequent factors have eigenvalues below 1. This first factor accounts for 40.61% of the variation in the items.

B Measure Validation

B.1 Informativeness

Table B.1: Two-sample Kolmogorov-Smirnov tests on RQ preceding marital decisions.

	d_0 = Full data		d_0 = Parent sample
	d_1 = Before separation	d_1 = Before marriage	d_1 = Before first child
$d_0 > d_1$	0.000	0.1741	0.2491
	(1.000)	(0.000)	(0.000)
$d_0 < d_1$	-0.2192	0.0000	-0.0000
	(0.000)	(1.000)	(1.000)
Combined	0.2192	0.1741	0.2489
	(0.000)	(0.000)	(0.000)

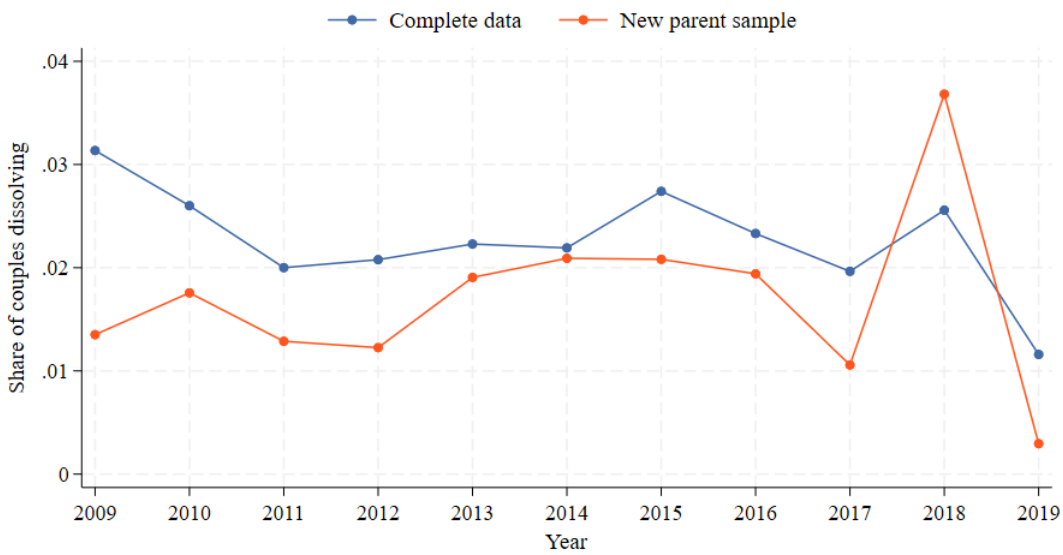
Notes: This table displays the results of two-sample Kolmogorov-Smirnov tests on different samples. The first two columns compare the distribution of RQ in the full data with the distribution the period before couple dissolution and marriage. The third column compares the overall distribution of the sample of parents with the distribution before childbirth within this sample. The null hypothesis is that the two samples compared are drawn from the same distribution. The reported coefficients are the resulting D-statistics and p-values, in parentheses.

Table B.2: RQ and marital transitions.

	Separation		Marriage	
	(1)	(2)	(3)	(4)
RQ	-0.00931***	-0.00862***	0.00320***	-0.000510
	(0.001)	(0.001)	(0.001)	(0.001)
Individual FE	✓	✓	✓	✓
Controls		✓		✓
R-squared	0.037	0.024	0.000	0.030
Observations	106826	106826	93854	84584

Notes: This table presents the results from regressing a binary variable indicating whether the period precedes couple dissolution – in columns (1) and (2) – and marriage – in columns (3) and (4) – on RQ. In columns (2) and (4) we control for age, sex, college education, employment status, log monthly income, presence of children, relationship tenure, marital status, area of residence, and period. Standard errors, clustered at the couple level, are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

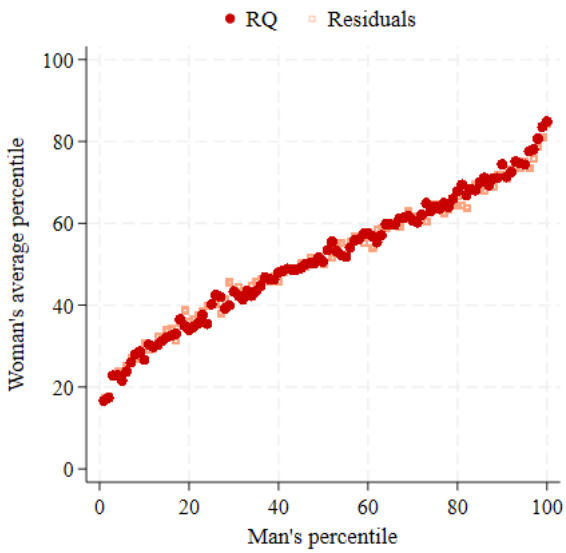
Figure B.1: Yearly couple dissolutions in data.



Notes: This figure plots the share of couples which dissolve out of all the couples observed each year in the complete data and in the new parent sample. According to Office for National Statistics (2022), on average 1.44% of the married couples in fertility ages (20-45 year-olds) living in England and Wales divorce every year, during the period 2009-2021.

B.2 Interpersonal comparability

Figure B.2: Rank-rank correlation of RQ across couple members.



Notes: This figure presents the rank-rank correlation of RQ between couple members in the full data. We plot the average RQ percentile rank of men against their female partners' RQ percentile rank. Each dot represents a percentile. Red markers denote RQ averages. Light markers correspond to residuals obtained regressing RQ on observable characteristics: age, educational attainment, marital status, and tenure.

Table B.3: Regression of woman RQ on man RQ.

	Woman RQ		
	(1)	(2)	(3)
Man RQ	0.613***	0.603***	0.594***
	(0.007)	(0.007)	(0.008)
Age × Tenure × Wave		✓	✓
Controls			✓
R-squared	0.320	0.330	0.334
Observations	42889	42889	39525

Notes: This table presents the results from regressing women's RQ on their (male) partners' RQ. Column (2) includes age, relationship tenure and survey wave dummies. Column (3) additionally controls for age, sex, college education, employment status, log monthly income, presence of children, relationship tenure, marital status, area of residence, and period. Standard errors, clustered at the couple level, are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.4: Couple correlation by item.

	Woman's response									
	(a) Subjective assessment					(b) Time use				
	(1) Consider splitting	(2) Regret marriage	(3) Quarrel	(4) Get on nerves	(5) Degree of happiness	(6) Work together	(7) Exchange ideas	(8) Calmly discuss	(9) Kiss	(10) Shared interests
Man's response	0.4095***	0.3115***	0.4586***	0.3930***	0.2811***	0.3230***	0.2714***	0.2822***	0.5326***	0.4378***
	(0.0105)	(0.0102)	(0.0068)	(0.0068)	(0.0064)	(0.0058)	(0.0060)	(0.0062)	(0.0059)	(0.0063)
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
R-squared	0.188	0.100	0.223	0.172	0.109	0.138	0.114	0.108	0.345	0.218
Observations	45356	45328	45323	45316	44948	45205	45156	45237	45184	36425

Notes: This table presents the results from regressing women's response to the items in Table 1 on their (male) partners' response to the same item. Columns (1) to (5) correspond to subjective assessment items, whereas columns (6) to (10) report correlations for time use items. Standard errors are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

C Additional Descriptives

Table C.1: Regression of RQ on individual, couple and partner characteristics, by gender.

	Women				Men			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(a) Individual characteristics								
College	0.0988***	0.0853***	0.0453**	-0.0384	0.1342***	0.1027***	0.0671***	0.0357
	(0.0174)	(0.0172)	(0.0205)	(0.0521)	(0.0174)	(0.0172)	(0.0212)	(0.0535)
Employed	0.1251***	0.1132***	0.0914***	0.0408	0.1410***	0.1074***	0.0841***	0.0289
	(0.0238)	(0.0236)	(0.0266)	(0.0251)	(0.0213)	(0.0213)	(0.0240)	(0.0228)
Log income	0.0239***	0.0257***	0.0245***	0.0015	-0.0027	-0.0069	-0.0010	-0.0051
	(0.0075)	(0.0074)	(0.0081)	(0.0070)	(0.0072)	(0.0071)	(0.0074)	(0.0064)
(b) Couple characteristics								
Married		0.2068***	0.2273***	0.0963***		0.2984***	0.2589***	0.0848**
		(0.0222)	(0.0251)	(0.0319)		(0.0230)	(0.0270)	(0.0350)
Children		-0.2277***	-0.2410***	-0.0545***		-0.2773***	-0.2793***	-0.0625***
		(0.0176)	(0.0189)	(0.0200)		(0.0190)	(0.0211)	(0.0217)
Urban		-0.0508***	-0.0495***	-0.0124		-0.0801***	-0.0761***	0.0013
		(0.0171)	(0.0182)	(0.0369)		(0.0178)	(0.0198)	(0.0398)
Age dif. ≥ 5		-0.0347**	-0.0030	0.0246		-0.0791***	-0.0274	0.1533
		(0.0169)	(0.0190)	(0.1487)		(0.0163)	(0.0197)	(0.1319)
(c) Partner characteristics								
College			0.0849***	0.0113			0.0813***	0.0615
			(0.0196)	(0.0454)			(0.0216)	(0.0560)
Employed			0.0382*	-0.0021			0.0371	0.0148
			(0.0220)	(0.0208)			(0.0268)	(0.0249)
Log income			0.0036	-0.0087			0.0340***	0.0103
			(0.0068)	(0.0058)			(0.0083)	(0.0072)
Age and wave	✓	✓	✓	✓	✓	✓	✓	✓
Tenure		✓	✓	✓		✓	✓	✓
Individual FE				✓				✓
R-squared	0.024	0.045	0.051	0.078	0.026	0.054	0.056	0.088
Observations	31769	31659	26253	26253	36848	36712	27907	27907

Notes: This table presents the results from regressing RQ on a set of individual, couple, and partner characteristics, separately for women – columns (1) to (4) – and men – columns (5) to (8). Columns (1) and (4) include individual characteristics: age, college education, employment, log monthly gross personal income, age and survey wave. In columns (2) and (6), we add couple characteristics: marital status, presence of children, urban residence, age difference and tenure. Columns (3) and (7) further include partner characteristics: college education, employment and log monthly income. Columns (4) and (8) add individual fixed effects. Standard errors clustered at the couple level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

C.1 Assortative matching on education

A large body of the family economics literature examines assortative matching, the tendency of individuals to pair with partners who have similar observable characteristics (Choo and Siow, 2006; Dupuy and Galichon, 2014; Chiappori et al., 2020; Blasutto, 2024). Assortative matching on education, in particular, is often considered to importantly influence couple formation and dissolution, as well as the way in which intra-couple decisions are made (Chiappori et al., 2017). In this line, Eckstein et al. (2019) consider assortative matching on education to be part of the match quality of a couple.

Our analysis in Table 4 provides evidence of assortative matching is positively associated with RQ, revealing a positive correlation between an individual's RQ and both their own and their partner's education. To explore this further, we categorize couples into four groups based on their educational attainment, following the classification of Costa and Kahn (2000). These categories are: low-power couples, where neither partner holds a college degree; part-power couples where only the woman is college educated; part-power couples where only the man is college educated; and high-power couples, in which both partners hold a college degree.

Table C.2 shows the distribution of these categories within our sample. Nearly 60% of the couples are low-power couples and around 17% are high-power, indicating that over three fourths of the couples in our sample have matching educational levels. Among part-power couples, those with a college educated woman slightly outnumber those with a college educated man, constituting 12% of the couples.

To study the association between sorting on education and RQ, we regress RQ on dummies for each couple type, with low-power couples as the baseline. We control for the rest of the characteristics in Table 4 and cluster standard errors at the couple level. The results, presented in Table C.3 (1), indicate that low-power couples have the lowest RQ levels. Both types of part-power couples exhibit higher RQ, with coefficients exceeding 0.04, although this is statistically significant only for part-power couples with college educated women. High-power couples report the highest RQ, with individuals in these couples showing an RQ that is 0.15 standard deviations higher than those in low-power couples. Only the coefficients corresponding to being married and having children are larger.

This result, combined with the findings in Table 4, where college education is associated with higher RQ in between-group comparisons but not in within-group comparisons, suggests that the positive association between education and RQ may reflect unobserved characteristics shared by college-educated individuals that contribute to better relationships. This aligns with the observations of Dupuy and Galichon (2014), who highlight the importance of shared personality traits in couple formation. Another possible explanation involves shared gender norms within education groups. Blasutto (2024) show that high-power couples tend to have larger gender wage gaps and greater household specialization, consistent with traditional gender roles. Similarly, Bertrand et al. (2015) find that adherence to traditional gender norms can strengthen marital stability.

We check for gender differences in this association, introducing an interaction term between gender and couple type. Table C.3 (2) presents these estimates. High-power couples continue to display significantly higher RQ values relative to low-power couples. The differences between part- and low-power couples narrow and become statistically insignificant. Notably, women in part-power couples where the man holds a college degree experience an 8.5% higher RQ compared to men in the same couple type.

Table C.2: Assortative matching on education in the data.

	Man's Education		
Woman's Education	No College	College	Total
No College	59.32	11.15	70.47
College	12.43	17.11	29.53
Total	71.75	28.25	100.00

Notes: This table presents the percentage of couples at any given combination of educational attainment of the partners – college or below college. Columns correspond to the educational level of men, while rows contain women's educational level.

Table C.3: Assortative matching on education and RQ.

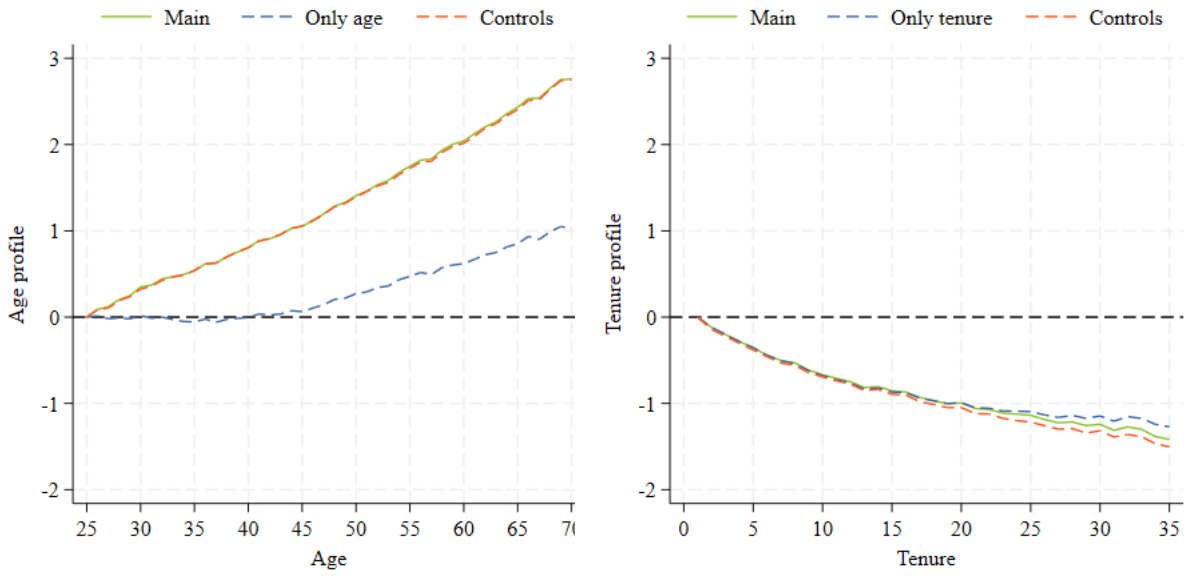
	(1)	(2)
Female	-0.092***	-0.112***
	(0.009)	(0.012)
Woman college	0.049*	0.028
	(0.022)	(0.024)
Man college	0.043	-0.000
	(0.026)	(0.029)
Both college	0.152***	0.139***
	(0.020)	(0.022)
Female × Woman college		0.042
		(0.024)
Female × Man college		0.085**
		(0.027)
Female × Both college		0.024
		(0.020)
Age × Tenure × Wave	✓	✓
Controls	✓	✓
Observations	54160	54160
R^2	0.054	0.055

Notes: This table presents the results from regressing RQ on education variables of both partners. Column (1) contains dummies for college education of the woman and the man, as well as their interaction. Column (2) additionally studies the different valuation of partners' education by gender, including the interaction between a female dummy and the education variables in column (1). Both columns control for own and partner's employment status and log monthly income, marital status, presence of children, urban residence and age difference, as well as age, relationship tenure and wave dummies. Standard errors clustered at the couple level in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

C.2 Conditional age and tenure profiles

Figure C.1: Age and tenure profiles of RQ.

(a) Age (b) Tenure



Notes: This figure plots the (a) age and (b) tenure profiles of RQ estimated in different ways. The green line corresponds to the estimates of Equation 1. The blue line estimates Equation 1 separately for age and tenure effects. The orange line adds a set of controls to Equation 1: gender, college education, employment, log monthly income, marital status, presence of children, urban residence and age difference between partners. The baseline for γ_a are 25 year-olds and the baseline for δ_d are one year long relationships.

D Duration

D.1 Describing RQ shocks

We define shocks to match quality based on period-to-period changes in RQ. Figure D.1 displays the distribution of these changes in our sample. We study two different shock sizes: small shocks, corresponding to a decrease in RQ of half standard deviation of the period-to-period change distribution, and big shocks, corresponding to a one standard deviation decrease. One standard deviation of the period-to-period change corresponds to 0.83 RQ units (standard deviations). As presented in Table D.1, the average size of a small shock is of about 1 RQ unit, whereas the average size of a big shock is of 1.4 RQ units.

Figure D.1: Period-to-period changes in RQ.

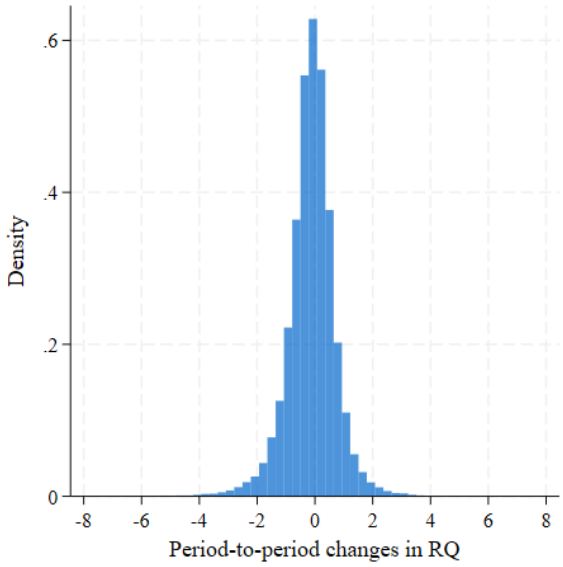


Table D.1: Average shock size in terms of standard deviations of RQ.

	(1) Mean	(2) Std dev.	(3) Min.	(4) Max.
Small shock (1/2 SD)	-1.011	-6.807	-0.415	0.654
Big shock (1 SD)	-1.446	-6.807	-0.830	0.693

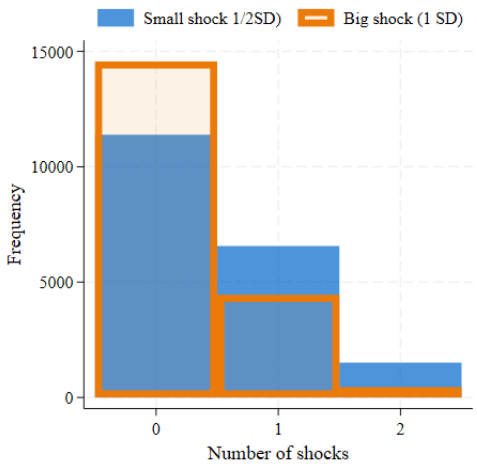
Notes:

Shocks at the couple level. We start by looking at the number of shocks that each couple experiences throughout the observation period.[19] Figure D.2 (a) displays the distribution of the

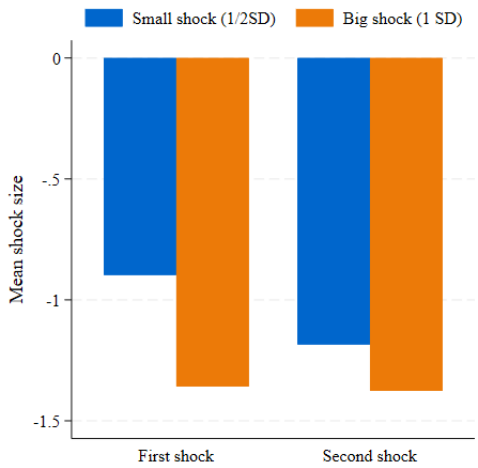
[19] We construct shocks at the couple level for this analysis. If both partners report a decrease in RQ, the size we consider is the larger one. While there are some exceptions, in most cases both partners experience simultaneous shocks. This is the case both for small and large shocks.

Figure D.2: Shock number and size at the couple level.

(a) Number of shocks per couple.



(b) Size of subsequent shocks.



total number of shocks experienced by couples. Around 62% of couples in our sample experience at least a small shock and 25% experience at least one big shock. On average, couples in our sample experience 0.5 small shocks and 0.3 big shocks throughout our observation period.

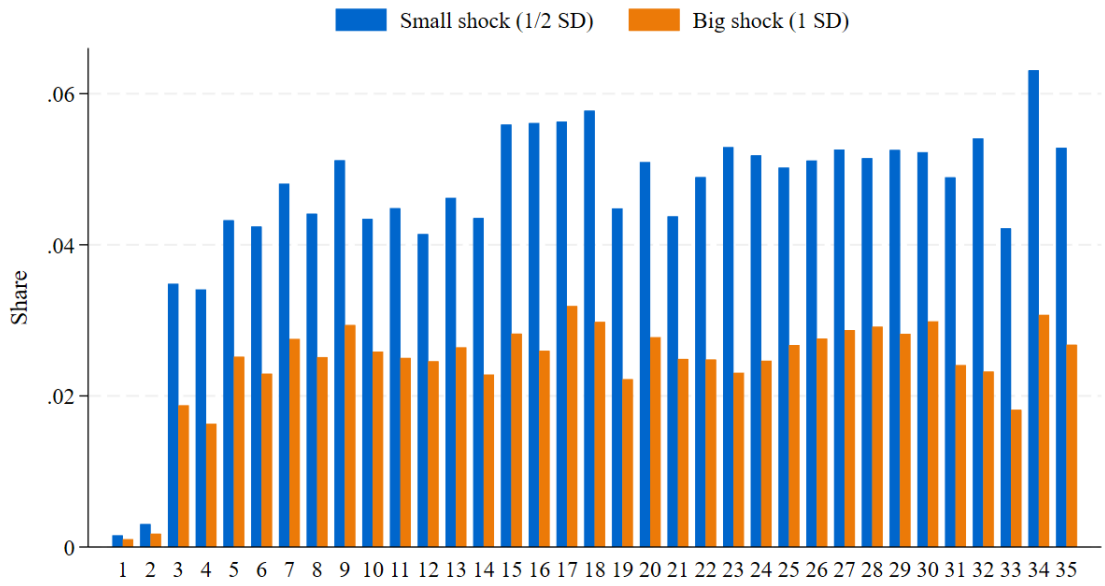
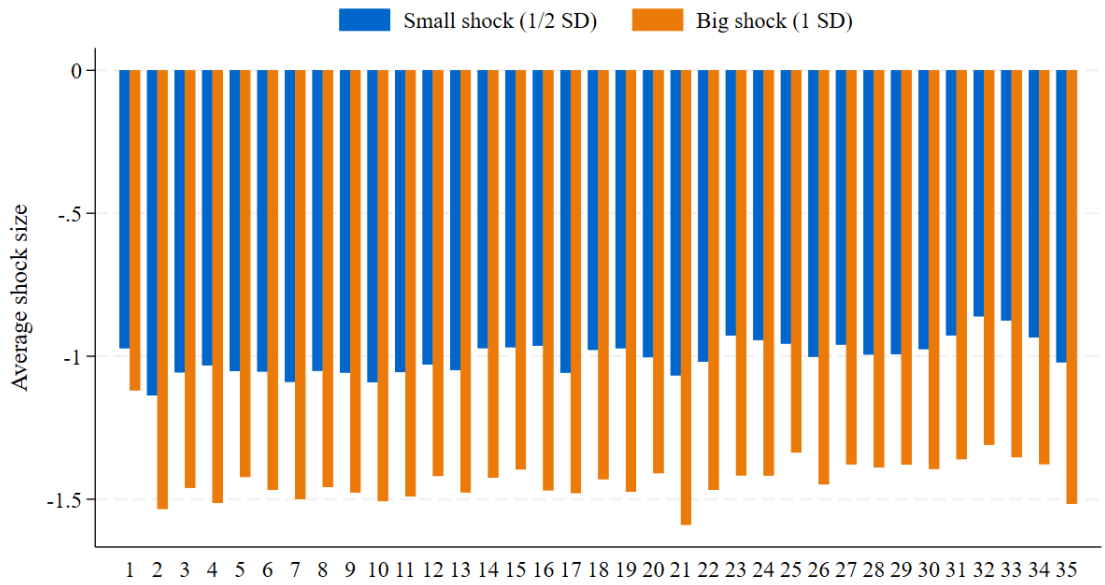
Figure D.2 (b) further studies the size of the subsequent shocks in terms of RQ units. The second shock is, on average, 0.2 standard deviations larger than the first one. Restricting to repeated big shocks, the size difference is substantially smaller but still existing. The average size of the first shock is computed using all individuals experiencing a shock. If we restrict to those experiencing two shocks this figure is virtually unchanged.

Shocks over tenure. Figure D.3 (a) displays the share of couples at each tenure that we observe experiencing a shock. Since our shocks are defined as period-to-period changes in RQ and our data is biannual, we cannot observe many shocks for early-stage relationships (those 2 years or shorter). The small share of those identified are couples who immediately experience a steep decline in RQ. Throughout the rest of relationship tenures, the share of couples experiencing shocks is rather uniform: around 4% in each tenure experience a small shock and about 2% suffer a large shock. There is no clear pattern in the propensity of shocks with tenure.

Figure D.3 (b) studies the size of shocks at different tenures. Shock sizes do not vary substantially with tenure, being in line with the average values outlined in Table D.1: around 1 RQ unit for small shocks and slightly below 1.5 RQ units for big shocks.

Figure D.3: Shock number and size at the couple level.

(a) Share of couples at each tenure experiencing a shock.



(b) Size of shocks at different tenures.